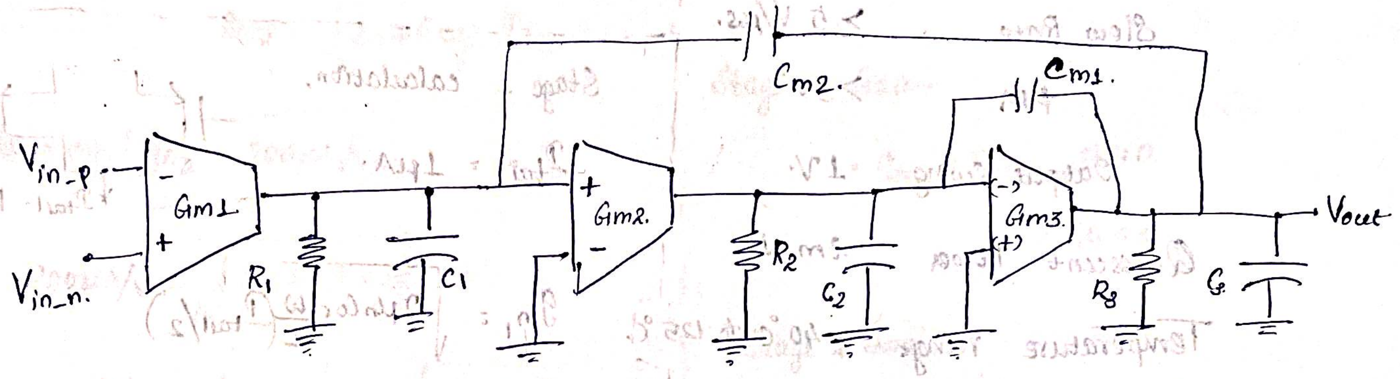
ECEN → 607 : Advanced Analog Circuit Tech.

a) Multi - Stage Op-Amp with Nested Miller Compensation.

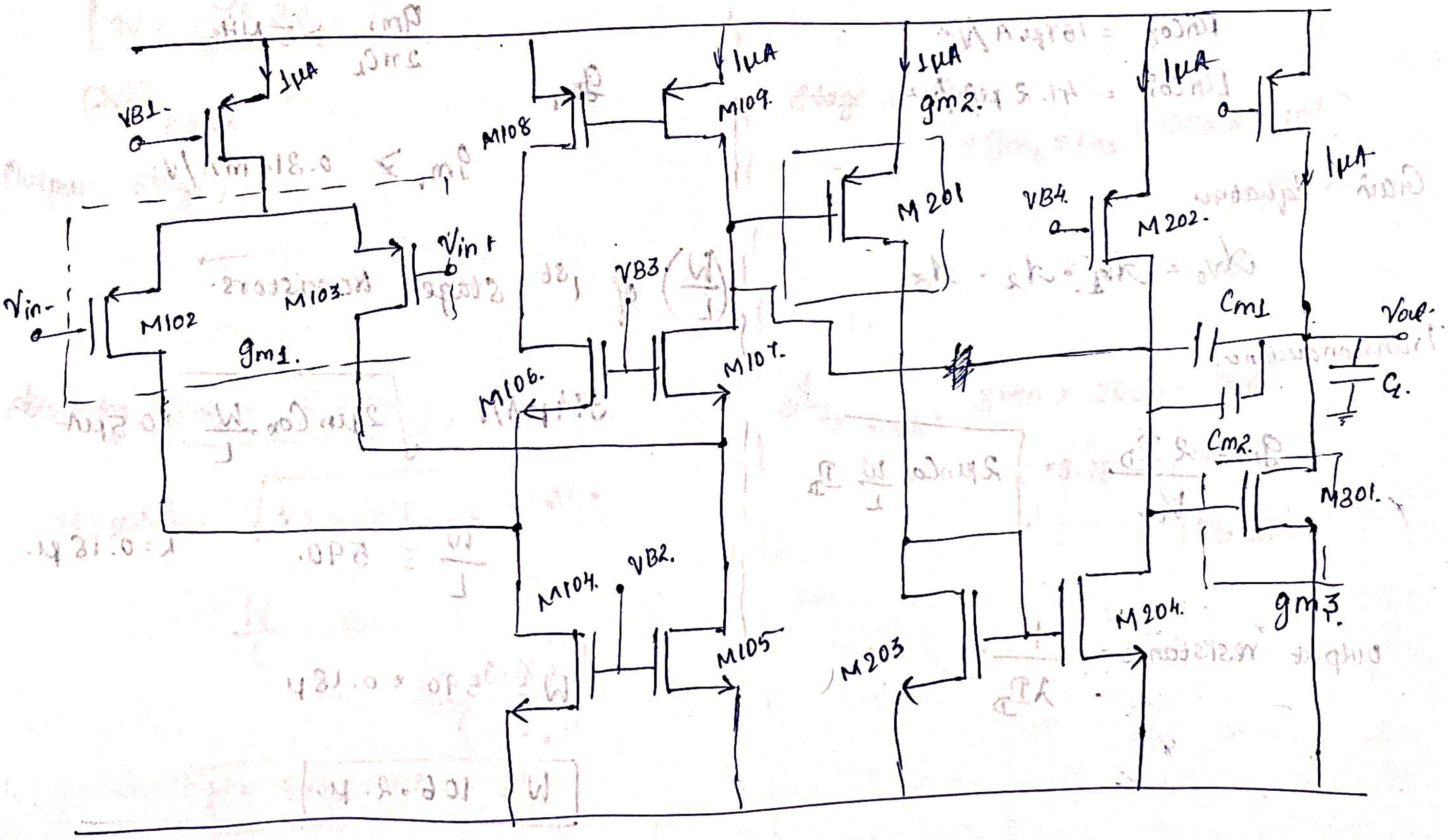
b) Using Cascoded Miller Compensation and Ahuja Compensation.

a) Design of 3 - stage Op-Amp.

Circuit Diagram:



The transistor architecture to be used :-



Design Specifications:-

Supply Voltage range	:	1.8 ± 5%.
Avo.	=	100 dB.
C load	=	10pF.
Unity Gain Frequency	:	> 5 MHz.
Slew Rate		> 5 V/μs.
PM		> 50°.
Output Swing		1V.
Quiescent Power		2mW.
Temperature range		−40°C to 125°C.

Process and Transistor Parameters:-

$\mu_n C_{ox} = 167 \mu A/V^2$.

$\mu_p C_{ox} = 41.2 \mu A/V^2$

Gain Equation

$$A_{v_o} = A_{v_1} \cdot A_2 \cdot A_3.$$

Transconductance.

$$g_m = \frac{2 I_D}{V_{ov}} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}$$

$$\text{output resistance} = \frac{1}{\lambda I_D}$$

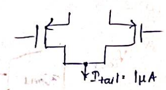
UGF

$= g_{m1}/2\pi C_L.$

Slew rate =, $SR = \dfrac{I_{stage\,2}}{C_L}$

Power Consumption : $P = V_{dd} \times I_{total}.$

Stage 1 calculation,



$I_{tail} = 1\mu A.$

$$g_{m_1} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} (I_{tail}/2)}$$

UGF > 5 MHz:

$$\frac{g_{m_1}}{2\pi C_L} > 5 MHz$$

$~~g_{m_1}~~$

$$g_{m_1} \geq 0.314 mA/V.$$

$\left(\frac{W}{L}\right)$ of 1st stage transistors.

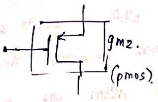
$$314 \mu A/V = \sqrt{2 \mu_n C_{ox} \frac{W}{L} \times 0.5\mu A}$$

$$\frac{W}{L} = 590. \qquad L = 0.18\mu.$$

$$W = 590 \times 0.18\mu.$$

$$\boxed{W = 106.2 \mu.}$$

Stage 2:



Same current through the branch,

$$I_{D2} = 1\mu A.$$

$$g_{m_2} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_{D2}}.$$

Assuming, $\boxed{g_{m_2} = 200\mu A/V.}$

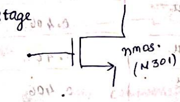
$$200\mu A/V = \sqrt{2 \times 41.2 \times \frac{W}{L} \times 1}$$

$$\frac{W}{L} = 424. \qquad \boxed{, L = 0.18\mu.} \xrightarrow{} \text{pmos}$$

$$\boxed{W = 76.32\mu.}$$

~~(W/L)~~ ~~nmos~~ =

Output stage



Assuming $\boxed{g_{m3} = 100\mu A/V.}$

$$100\mu A/V = \sqrt{2 \times 167 \times \frac{W}{L} \times 1.}$$

$$\frac{W}{L} = 30.$$

$\boxed{L = 0.18\mu.}$

$\boxed{W = 5.4\mu.}$ nmos

$$\boxed{\left(W/L\right)_{pmos} = 424.}$$

Gain Calculation.

$$r_0 = 1/\lambda I_D.$$

Assuming $\lambda = 0.1 V^{-1}$

$$r_0 = \frac{1}{(0.1)(1 \times 10^{-6})} = 10^7 \Omega$$

Stage 1 Gain =

$$g_{m_1} \times r_{01} = 3140.$$

$$= 70 dB.$$

Stage 2 Gain,

$$= g_{m_2} \times r_{02} = 2000.$$

$$= 66 dB.$$

Stage 3 Gain.

$$= g_{m_3} \times r_{03} = 100\mu A \times 10^7$$

$$= 1000.$$

$$= 60 dB.$$

$$A_{v_o, total} = 3140 \times 2000 \times 1000$$

$$= 6.28 \times 10^9$$

$$= 196 dB$$

Current in Each Branch:

Input pair ; $I_{tail} = 1\mu A$;

$I_{D1} = I_{D2}$ = Each input transistor $= 0.5\mu A$

Slew Rate Check:

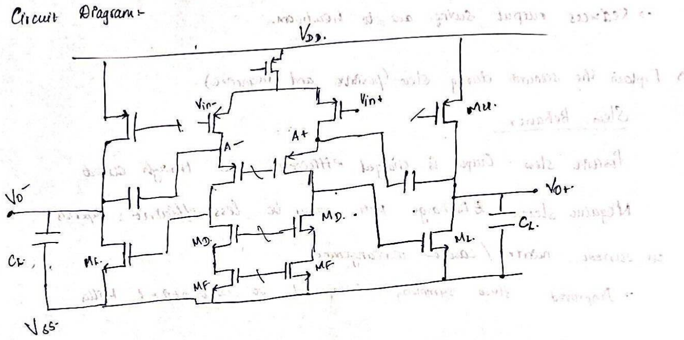
$$SR = \frac{I_{out}}{C_L} = \frac{1\times10^{-6}}{10\times10^{-12}} = 100\ V/\mu s$$

Transistor	TYPE	FUNCTION / STAGE	CURRENT (µA)	W/L ; L = 0.18µ	App. VGS
M101	PMOS	Bias	1	76.32µ	0.379
M102	PMOS	Diff Pair (in−)	0.5	76.32µ	0.385
M103	PMOS	Diff Pair (in+)	0.5	76.32µ	0.385
M104	NMOS	Bias	1	5.40µ	0.406
M105	NMOS	Bias	1	5.40µ	0.406
M106	NMOS	Cascode	1	5.40µ	0.406
M107	NMOS	Cascode	1	5.40µ	0.406
M108	PMOS	Cascode	1	76.32µ	0.379
M109	PMOS	Cascode	1	76.32µ	0.379
M201	PMOS	Stage 2	1	76.32µ	0.379
M202	NMOS	Stage 2	1	106.20µ	0.464
M203	NMOS	Bias	1	5.4µ	0.406
M204	NMOS	Bias	1	5.4µ	0.406
M301	PMOS	Output Stage	1	5.4µ	0.406
M302	PMOS	Output Stage	1	76.32µ	0.379

2). Figure 1.2 shows a fully differential two-stage op-amp (Stage 1 is telescopic, stage 2 is common source),

Usually, the miller caps, (C_c) are connected to nodes B+ and B−,
In this figure, it is connected to the cascoding node (A and A+) to mitigate the RHP zero effect and is commonly known as Cascoded miller compensation :-

Circuit Diagram :-



(a) How does the compensation work here ? Show relevant equation.

Cascoded Miller caps connect to low-impedance cascode (A, A+).

The Miller capacitor current :

$i_c = sC_m(V_{in} - V_{out})$ produces a right-half plane (RHP) zero

Ahuja's compensation instead injects Miller current at the cascode node : $i_c = sC_m V_{in}$.

This eliminates the $-sC_mV_{out}$ term, removing the RHP zero,

(b) Advantages and Disadvantages.

Advantages:-

→ Eliminates RHP. zero, enhancing stability

→ Improves phase margin and settling.

Disadvantages:-

→ More complex (needs cascode structure)

→ Reduces output swing due to headroom.

(c) Explain the scenario during slew (positive and negative)

<u>Slew Behavior:-</u>

Positive slew : Output is charged efficiently ~~via~~ through cascode.

Negative slew : Discharge path, may be less efficient; depends on current mirror / cascode arrangements.

→ Improved slew symmetry compared to conventional Miller.